Full Experiment Results

Supplement to the CS seminar paper (CS_seminar_IFTY)

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1 What this document is

The 4-page paper can only show summaries. This document lists every result of every experiment that I ran on Kaggle for the new stage of E1, where all three modules are ML components. All tables are generated directly from the result files that the Kaggle kernels wrote (the generator is `experiments/E1_new/analysis/make_results_doc.py`), so no number was copied by hand.

Notation used everywhere:

- C_1 perception (YOLOv11n), C_2 prediction (AutoBot-Ego style), C_3 planning (PlanT style).
- **old** C_1: C_1 trained on Boston 1. Two copies, seed 0 (s0) and seed 1 (s1).
- **update**: retraining an old C_1 on Singapore 1. Update names read `b0 f50 e10 s2` = base old C_1 seed 0, 50 % of the Singapore images, 10 epochs, update seed 2.
- **isolated training** (iso): C_2, C_3 trained on the ground truth. **old-pipeline training** (op): trained on the output of the old C_1. **new-pipeline training** (np): trained on the output of the new C_1.
- C_3 comes in two variants: **ego** (gets the last 1.5 s of the ego path, used in the paper) and **noego** (no ego input, failed its baseline, kept here for completeness).

2 Definitions and formulas

- **Accuracy gain of** C_1: $\delta_1 = \mathrm{mAP50}(C_1^{\mathrm{new}}) - \mathrm{mAP50}(C_1^{\mathrm{old}})$ on Singapore 2.
- **Prediction error** (minADE over $M = 6$ modes, $T = 6$ steps of 0.5 s, m_t = valid step):

$$\mathrm{minADE} = \frac{1}{N}\sum_{n=1}^{N} \min_{k=1..M} \frac{\sum_t m_t \, \|\hat{p}_{n,t}^{(k)} - p_{n,t}\|_2}{\sum_t m_t}$$

  over the objects found by C_1 that are matched one to one (Hungarian, same class, radius $\max(2, 0.15 \cdot \mathrm{range})$ m) to a ground-truth object.
- **Planning error**: $\mathrm{L2@}t = \frac{1}{A}\sum_a \|\hat{e}_a(t) - e_a(t)\|_2$ for $t = 1, 2, 3$ s over the $A = 1{,}720$ test samples (e = human ego path).
- **Collision rate**: fraction of test samples in which an oriented box of the ego vehicle (4.08 m × 1.73 m) on the planned path overlaps a ground-truth object box at any of the 6 future steps.
- **Object use**: mean distance between the planned 3 s position with and without the objects in the input of C_3.
- $\Delta_2 = \mathrm{minADE}_{\text{new } C_1} - \mathrm{minADE}_{\text{old } C_1}$, $\quad \Delta_3 = \mathrm{L2@3s}_{\text{new } C_1} - \mathrm{L2@3s}_{\text{old } C_1}$ (same C_2, C_3).
- **Entangled enhancement**: $\delta_1 > 0$ and ($\Delta_2 > 0$ or $\Delta_3 > 0$).
- **Random variation** (level 2 threshold): $\tau = \max|\Delta|$ when the old C_1 is swapped for the other old C_1 (both bases, both trainings): $\tau_2 = 0.112$ m, $\tau_3 = 0.018$ m.
- **Bootstrap** (level 3): 2,000 resamples of the 50 test scenes with replacement; per resample $\Delta = \bar{e}_{\mathrm{new}} - \bar{e}_{\mathrm{old}}$ on the same scenes; harm = 2.5th percentile > 0.
- **Jensen–Shannon divergence** (base 2): $\mathrm{JS}(P,Q) = \frac{1}{2}\mathrm{KL}(P\|M) + \frac{1}{2}\mathrm{KL}(Q\|M)$, $M = \frac{1}{2}(P+Q)$; 0 = equal, 1 = disjoint.
- **Wasserstein distance**: $W_1(P,Q) = \int |F_P(x) - F_Q(x)|\,dx$. **KS statistic**: $\sup_x |F_P(x) - F_Q(x)|$.
- **Only-one-version fraction**: share of objects found by only one of the two C_1 on the same image (same class, within 2 m counts as the same object).
- **Spearman rank correlation** ρ and **AUROC** = P(measure of a harmful update > measure of a harmless update), harmful = $\Delta_2 > 0$ after old-pipeline training.

3 Data and compute

Table 1: The 350-scene subset (full-size CAM_FRONT keyframes, 4.645 GB in total, 0 missing images)

Part	Scenes	Keyframes	Images with labels	Used for
Boston 1 (boston_c1)	100	4,008	3,923	training the old C_1
Boston 2 (boston_c23)	100	4,046	–	training C_2, C_3 (86 train, 14 validation)
Singapore 1 (singapore_update)	100	3,997	3,575	updates of C_1
Singapore 2 (singapore_test)	50	2,020	1,805	testing (1,720 test samples)

Table 2: Kaggle kernels and run times

Stage	Kernel	GPU	Run time
Subset	ifty1011/labsd-e1-subset350	CPU	434.5 s
Stage 0 (old and new C_1)	ifty1011/labsd-e1-stage0-c1	T4	2,546 s
Stage 1 data export	ifty1011/labsd-e1-stage1-data	T4	953 s
Stage 1b (C_2, C_3)	ifty1011/labsd-e1-stage1b	T4	2,365 s
Stage 2 campaign, base s0	ifty1011/labsd-e1-stage2-campaign	T4	9,096 s
Stage 2 campaign, base s1	rai73416/labsd-e1-stage2-campaign	T4	8,002 s
Stage 3 bootstrap, base s0	ifty1011/labsd-e1-stage3-bootstrap	CPU	421.5 s
Stage 3 bootstrap, base s1	rai73416/labsd-e1-stage3-bootstrap	CPU	423.0 s

4 Stage 0: old and new C_1

YOLOv11n, 10 epochs, image size 640, batch 8, checkpoint `last.pt`, never selected on the test set. Training times: old s0 579 s, new s0 493 s, old s1 535 s, new s1 487 s. δ_1 (Stage 0 updates): seed 0 +0.0750, seed 1

Table 3: The four Stage 0 models on Singapore 2

Model	Init	Trained on	mAP50	mAP50-95	Precision	Recall	Detections
c1 old s0	yolo11n.pt	boston c1	0.1086	0.0483	0.363	0.105	3,563
c1 new s0	c1 old s0	singapore update	0.1836	0.0833	0.341	0.194	3,603
c1 old s1	yolo11n.pt	boston c1	0.1173	0.0543	0.383	0.116	3,705
c1 new s1	c1 old s1	singapore update	0.1850	0.0855	0.364	0.185	3,409

+0.0677. Seed gap of the old C_1 (mAP50): 0.0087.

Table 4: First look at the output change of C_1 (no ground truth), 1335 images with any detection

Measure	old s0 vs old s1	new s0 vs new s1	old s0 vs new s0	old s1 vs new s1
objects per image JS	0.0065	0.0052	0.0068	0.0140
mean objects per image	2.67/2.78	2.70/2.55	2.67/2.70	2.78/2.55
type mix JS	0.0056	0.0027	0.0372	0.0367
confidence W	0.0070	0.0036	0.0356	0.0350
confidence KS	0.0321	0.0298	0.0910	0.0753
forward position W m	9.2614	4.1499	12.7059	11.0180
lateral position W m	1.4852	5.8968	2.5465	5.2604
only one version fraction	0.3990	0.3682	0.5091	0.5105

5 Stage 1 data export

Ground truth and the detections of the four Stage 0 models, per keyframe, for Boston 2 (4,046 keyframes) and Singapore 2 (2,020 keyframes). The Singapore 2 counts are identical to Stage 0 (consistency check passed).

Table 5: Number of detections per model

Split	old s0	new s0	old s1	new s1
Boston 2	25,286	20,511	26,813	23,663
Singapore 2	3,563	3,603	3,705	3,409

6 Stage 1: first ML version of C_2 and C_3 (failed the baselines)

First version: scene-level C2 (no target-centred view), C3 without any ego input, 40/60 epochs, no validation set, no augmentation. Both modules learned Boston but did not beat the baselines on Singapore 2 (stay-still 4.48 m for C_2, mean path per command 9.23 m for C_3). This version was replaced by Stage 1b.

Table 6: Stage 1 results on Singapore 2 (C3 = noego)

C1 seed	Training	C2 iso	C1 used	C2 pipe	C3 L2@1s	L2@2s	L2@3s	Collision
0	isolated	4.681	old s0	5.993	3.97	7.99	11.911	0.195
0	isolated	4.681	new s0	5.832	3.87	7.79	11.630	0.178
0	isolated	4.681	old s1	6.111	3.81	7.66	11.434	0.193
0	old-pipeline	6.452	old s0	6.150	4.02	8.05	11.990	0.191
0	old-pipeline	6.452	new s0	5.843	4.00	8.00	11.890	0.184
0	old-pipeline	6.452	old s1	6.131	3.93	7.87	11.737	0.181
0	new-pipeline	7.170	new s0	5.547	3.44	6.95	10.500	0.187
1	isolated	4.715	old s1	6.045	3.77	7.57	11.323	0.196
1	isolated	4.715	new s1	5.823	3.87	7.76	11.589	0.188
1	isolated	4.715	old s0	5.910	3.94	7.90	11.799	0.204
1	old-pipeline	6.580	old s1	6.186	3.61	7.31	10.980	0.178
1	old-pipeline	6.580	new s1	6.091	3.58	7.26	10.887	0.172
1	old-pipeline	6.580	old s0	6.250	3.64	7.38	11.061	0.183
1	new-pipeline	6.995	new s1	5.684	3.34	6.78	10.283	0.185

7 Stage 1b: final C_2 and C_3

Target-centred C_2 (736,799 parameters), C_3 (619,010 parameters, ego and noego variants), early stopping on the 14 Boston validation scenes, mirror and rotation augmentation. Baselines on Singapore 2: C_2 stay-still 4.477 m; C_3 mean path per command L2@1/2/3s = 3.10/6.17/9.251 m, collision 0.177. Constant-velocity reference with ground-truth identities: 1.008 m (test), 0.579 m (Boston validation).

Table 7: Best validation error (Boston 2 validation scenes)

Model	best val minADE (m)	best val L2@3s (m)
p1 isolated seed0 c2	0.364	–
p1 isolated seed0 c3 noego	–	7.589
p1 isolated seed0 c3 ego	–	2.759
p1 old pipeline c1s0 seed0 c2	2.164	–
p1 old pipeline c1s0 seed0 c3 noego	–	8.063
p1 old pipeline c1s0 seed0 c3 ego	–	2.848
p1 new pipeline c1s0 seed0 c2	2.202	–
p1 new pipeline c1s0 seed0 c3 noego	–	7.267
p1 new pipeline c1s0 seed0 c3 ego	–	2.725
p1 old pipeline c1s1 seed0 c2	2.175	–
p1 old pipeline c1s1 seed0 c3 noego	–	7.878
p1 old pipeline c1s1 seed0 c3 ego	–	2.840
p1 new pipeline c1s1 seed0 c2	2.319	–
p1 new pipeline c1s1 seed0 c3 noego	–	7.521
p1 new pipeline c1s1 seed0 c3 ego	–	2.724

Table 8: Stage 1b results on Singapore 2 (1,720 test samples)

C1 seed	Training	C3	C1 used	C2 minADE	scored	stay-still	L2@1s	L2@2s	L2@3s	Collision	Obj. use	iso L2@3s	iso obj. use
0	isolated	ego	old s0	2.887	1119	5.676	1.00	2.33	3.989	0.076	0.73	3.821	1.51
0	isolated	ego	new s0	2.740	1101	5.390	1.01	2.35	4.020	0.075	0.81	3.821	1.51
0	isolated	ego	old s1	2.946	1118	5.853	1.00	2.34	4.007	0.076	0.78	3.821	1.51
0	isolated	noego	old s0	2.887	1119	5.676	4.26	8.49	12.548	0.194	1.06	11.952	2.98
0	isolated	noego	new s0	2.740	1101	5.390	4.19	8.36	12.350	0.176	1.19	11.952	2.98
0	isolated	noego	old s1	2.946	1118	5.853	4.13	8.24	12.203	0.174	1.16	11.952	2.98
0	old-pipeline	ego	old s0	2.615	1119	5.676	1.06	2.45	4.163	0.079	0.91	3.972	1.34
0	old-pipeline	ego	new s0	2.748	1101	5.390	1.06	2.44	4.143	0.074	0.89	3.972	1.34
0	old-pipeline	ego	old s1	2.727	1118	5.853	1.05	2.43	4.150	0.079	0.95	3.972	1.34
0	old-pipeline	noego	old s0	2.615	1119	5.676	4.05	8.09	11.975	0.189	3.65	11.026	4.57
0	old-pipeline	noego	new s0	2.748	1101	5.390	3.97	7.91	11.692	0.176	3.64	11.026	4.57
0	old-pipeline	noego	old s1	2.727	1118	5.853	3.94	7.85	11.619	0.177	3.82	11.026	4.57
0	new-pipeline	ego	new s0	2.389	1101	5.390	0.91	2.14	3.705	0.060	0.47	3.769	0.74
0	new-pipeline	noego	new s0	2.389	1101	5.390	3.60	7.18	10.684	0.182	4.04	11.236	4.53
1	isolated	ego	old s1	2.946	1118	5.853	1.00	2.34	4.007	0.076	0.78	3.821	1.51
1	isolated	ego	new s1	2.775	1132	5.553	1.01	2.36	4.028	0.076	0.79	3.821	1.51
1	isolated	ego	old s0	2.887	1119	5.676	1.00	2.33	3.989	0.076	0.73	3.821	1.51
1	isolated	noego	old s1	2.946	1118	5.853	4.13	8.24	12.203	0.174	1.16	11.952	2.98
1	isolated	noego	new s1	2.775	1132	5.553	4.20	8.37	12.380	0.180	1.01	11.952	2.98
1	isolated	noego	old s0	2.887	1119	5.676	4.26	8.49	12.548	0.194	1.06	11.952	2.98
1	old-pipeline	ego	old s1	2.660	1118	5.853	1.01	2.35	4.000	0.069	1.24	3.867	1.65
1	old-pipeline	ego	new s1	2.606	1132	5.553	1.01	2.34	4.000	0.070	1.13	3.867	1.65
1	old-pipeline	ego	old s0	2.562	1119	5.676	1.01	2.34	4.002	0.072	1.17	3.867	1.65
1	old-pipeline	noego	old s1	2.660	1118	5.853	3.43	6.91	10.370	0.156	3.28	9.255	6.19
1	old-pipeline	noego	new s1	2.606	1132	5.553	3.48	7.01	10.518	0.155	3.06	9.255	6.19
1	old-pipeline	noego	old s0	2.562	1119	5.676	3.57	7.21	10.824	0.170	3.32	9.255	6.19
1	new-pipeline	ego	new s1	2.446	1132	5.553	0.88	2.11	3.697	0.048	0.53	3.730	0.84
1	new-pipeline	noego	new s1	2.446	1132	5.553	3.30	6.61	9.959	0.168	2.64	10.601	4.07

Table 9: Stage 1b deltas (old $C_1 \rightarrow$ Stage 0 new C_1) and seed-swap noise

C1 seed	Training	Δ_2	noise Δ_2	Δ_3 ego	noise ego	Δ_3 noego	noise noego
0	isolated	−0.147	+0.059	+0.031	+0.018	−0.197	−0.345
0	old-pipeline	+0.133	+0.112	−0.020	−0.013	−0.282	−0.355
1	isolated	−0.170	−0.059	+0.021	−0.018	+0.177	+0.345
1	old-pipeline	−0.054	−0.098	−0.001	+0.001	+0.148	+0.454

8 Stage 2: the 30-update campaign

Each base old C_1 is updated with 15 settings: data $\{100, 50, 25\}\,\%\times$ seeds $\{1, 2, 3\}$ at 10 epochs, and epochs $\{5, 20\}\times$ seeds $\{1, 2, 3\}$ at $100\,\%$. C_2 and C_3 are the saved Stage 1b models and are never retrained. Base old C_1: s0 mAP50 = 0.1086, s1 mAP50 = 0.1173.

8.1 The updated C_1 models

Table 10: All 30 updated C_1 on Singapore 2

#	Update	Data	Epochs	Seed	Train images	mAP50	mAP50-95	Precision	Recall
1	b0 f100 e10 s1	100%	10	1	3,575	0.1860	0.0830	0.314	0.217
2	b0 f100 e10 s2	100%	10	2	3,575	0.2045	0.0907	0.397	0.219
3	b0 f100 e10 s3	100%	10	3	3,575	0.1802	0.0818	0.355	0.195
4	b0 f50 e10 s1	50%	10	1	1,788	0.1817	0.0833	0.393	0.203
5	b0 f50 e10 s2	50%	10	2	1,788	0.1852	0.0868	0.318	0.212
6	b0 f50 e10 s3	50%	10	3	1,788	0.1733	0.0820	0.316	0.188
7	b0 f25 e10 s1	25%	10	1	894	0.1729	0.0816	0.312	0.188
8	b0 f25 e10 s2	25%	10	2	894	0.1679	0.0765	0.275	0.198
9	b0 f25 e10 s3	25%	10	3	894	0.1758	0.0803	0.296	0.222
10	b0 f100 e5 s1	100%	5	1	3,575	0.2031	0.0942	0.322	0.239
11	b0 f100 e5 s2	100%	5	2	3,575	0.2095	0.0965	0.360	0.245
12	b0 f100 e5 s3	100%	5	3	3,575	0.2176	0.0956	0.370	0.242
13	b0 f100 e20 s1	100%	20	1	3,575	0.1779	0.0828	0.327	0.200
14	b0 f100 e20 s2	100%	20	2	3,575	0.1881	0.0842	0.378	0.212
15	b0 f100 e20 s3	100%	20	3	3,575	0.2112	0.0924	0.416	0.224
16	b1 f100 e10 s1	100%	10	1	3,575	0.1850	0.0855	0.364	0.185
17	b1 f100 e10 s2	100%	10	2	3,575	0.1838	0.0857	0.303	0.213
18	b1 f100 e10 s3	100%	10	3	3,575	0.1821	0.0839	0.336	0.204
19	b1 f50 e10 s1	50%	10	1	1,788	0.1826	0.0857	0.370	0.193
20	b1 f50 e10 s2	50%	10	2	1,788	0.1840	0.0832	0.339	0.218
21	b1 f50 e10 s3	50%	10	3	1,788	0.1959	0.0904	0.345	0.217
22	b1 f25 e10 s1	25%	10	1	894	0.1656	0.0770	0.316	0.184
23	b1 f25 e10 s2	25%	10	2	894	0.1951	0.0864	0.383	0.209
24	b1 f25 e10 s3	25%	10	3	894	0.1804	0.0853	0.314	0.205
25	b1 f100 e5 s1	100%	5	1	3,575	0.2031	0.0948	0.353	0.240
26	b1 f100 e5 s2	100%	5	2	3,575	0.2353	0.1089	0.455	0.248
27	b1 f100 e5 s3	100%	5	3	3,575	0.2324	0.1071	0.410	0.259
28	b1 f100 e20 s1	100%	20	1	3,575	0.2196	0.0949	0.434	0.230
29	b1 f100 e20 s2	100%	20	2	3,575	0.1884	0.0859	0.368	0.191
30	b1 f100 e20 s3	100%	20	3	3,575	0.1881	0.0840	0.401	0.193

8.2 Effect on C_2 and C_3, old-pipeline training

Table 11: Old-pipeline training: errors with the old C_1 (base) and with the updated C_1 (new), in metres

#	Update	δ_1	C2 base	C2 new	Δ_2	scored	ego base	ego new	Δ_3 ego	Δcoll. ego	obj. use	noego base	noego new	Δ_3 noego
1	b0 f100 e10 s1	+0.0773	2.615	2.664	+0.049	1199	4.163	4.136	−0.027	−0.003	0.90	11.975	11.863	−0.112
2	b0 f100 e10 s2	+0.0958	2.615	2.677	+0.062	1049	4.163	4.189	+0.026	−0.006	0.80	11.975	12.027	+0.053
3	b0 f100 e10 s3	+0.0715	2.615	2.754	+0.139	1094	4.163	4.183	+0.020	−0.006	0.86	11.975	11.904	−0.071
4	b0 f50 e10 s1	+0.0731	2.615	2.799	+0.184	1183	4.163	4.151	−0.012	−0.005	0.90	11.975	11.808	−0.166
5	b0 f50 e10 s2	+0.0765	2.615	2.807	+0.192	1318	4.163	4.194	+0.031	−0.001	0.90	11.975	11.850	−0.124
6	b0 f50 e10 s3	+0.0647	2.615	2.713	+0.098	1055	4.163	4.202	+0.040	−0.006	0.80	11.975	11.995	+0.020
7	b0 f25 e10 s1	+0.0643	2.615	2.848	+0.233	1230	4.163	4.133	−0.030	−0.006	0.91	11.975	11.677	−0.298
8	b0 f25 e10 s2	+0.0593	2.615	2.899	+0.284	1284	4.163	4.121	−0.042	−0.006	0.97	11.975	11.517	−0.458
9	b0 f25 e10 s3	+0.0672	2.615	2.772	+0.157	1237	4.163	4.117	−0.046	−0.001	1.03	11.975	11.372	−0.603
10	b0 f100 e5 s1	+0.0944	2.615	2.797	+0.182	1315	4.163	4.139	−0.024	−0.003	0.94	11.975	11.708	−0.266
11	b0 f100 e5 s2	+0.1008	2.615	2.900	+0.284	1279	4.163	4.157	−0.006	−0.005	0.90	11.975	11.841	−0.134
12	b0 f100 e5 s3	+0.1089	2.615	2.882	+0.266	1368	4.163	4.120	−0.042	−0.002	0.98	11.975	11.576	−0.399
13	b0 f100 e20 s1	+0.0693	2.615	2.722	+0.107	1024	4.163	4.163	−0.000	−0.006	0.83	11.975	11.884	−0.091
14	b0 f100 e20 s2	+0.0795	2.615	2.742	+0.127	989	4.163	4.188	+0.025	−0.004	0.80	11.975	11.883	−0.092
15	b0 f100 e20 s3	+0.1026	2.615	2.730	+0.115	1127	4.163	4.164	+0.001	−0.006	0.86	11.975	11.870	−0.105
16	b1 f100 e10 s1	+0.0677	2.660	2.606	−0.054	1132	4.000	4.000	−0.001	+0.001	1.13	10.370	10.518	+0.148
17	b1 f100 e10 s2	+0.0665	2.660	2.579	−0.081	1128	4.000	4.026	+0.026	−0.001	1.10	10.370	10.652	+0.282
18	b1 f100 e10 s3	+0.0648	2.660	2.473	−0.187	1108	4.000	4.023	+0.023	−0.001	1.18	10.370	10.686	+0.316
19	b1 f50 e10 s1	+0.0653	2.660	2.630	−0.030	1097	4.000	4.072	+0.072	+0.002	1.10	10.370	10.832	+0.462
20	b1 f50 e10 s2	+0.0667	2.660	2.525	−0.135	1061	4.000	4.054	+0.054	+0.000	1.07	10.370	10.741	+0.371
21	b1 f50 e10 s3	+0.0787	2.660	2.571	−0.089	1179	4.000	4.037	+0.037	+0.001	1.15	10.370	10.675	+0.304
22	b1 f25 e10 s1	+0.0483	2.660	2.750	+0.090	1127	4.000	3.999	−0.001	−0.001	1.18	10.370	10.439	+0.069
23	b1 f25 e10 s2	+0.0778	2.660	2.741	+0.081	1317	4.000	3.959	−0.042	+0.004	1.36	10.370	10.300	−0.070
24	b1 f25 e10 s3	+0.0631	2.660	2.742	+0.082	1292	4.000	4.011	+0.010	+0.001	1.21	10.370	10.460	+0.089
25	b1 f100 e5 s1	+0.0858	2.660	2.635	−0.025	1048	4.000	4.033	+0.033	−0.001	1.06	10.370	10.651	+0.281
26	b1 f100 e5 s2	+0.1180	2.660	2.699	+0.040	1204	4.000	4.004	+0.003	+0.001	1.19	10.370	10.746	+0.376
27	b1 f100 e5 s3	+0.1151	2.660	2.712	+0.052	1322	4.000	4.015	+0.015	−0.001	1.20	10.370	10.594	+0.223
28	b1 f100 e20 s1	+0.1023	2.660	2.663	+0.003	1102	4.000	4.050	+0.049	+0.001	1.05	10.370	10.717	+0.347
29	b1 f100 e20 s2	+0.0712	2.660	2.725	+0.065	1110	4.000	4.021	+0.021	−0.001	1.11	10.370	10.674	+0.303
30	b1 f100 e20 s3	+0.0708	2.660	2.733	+0.073	1101	4.000	4.018	+0.018	+0.000	1.11	10.370	10.675	+0.305
noise	old s0→old s1	–	2.615	2.727	+0.112	1118	4.163	4.150	−0.013	+0.000	0.95	11.975	11.619	−0.355
noise	old s1→old s0	–	2.660	2.562	−0.098	1119	4.000	4.002	+0.001	+0.003	1.17	10.370	10.824	+0.454

8.3 Effect on C_2 and C_3, isolated training

Table 12: Isolated training: errors with the old C_1 (base) and with the updated C_1 (new), in metres

#	Update	δ_1	C2 base	C2 new	Δ_2	scored	ego base	ego new	Δ_3 ego	Δcoll. ego	obj. use	noego base	noego new	Δ_3 noego
1	b0 f100 e10 s1	+0.0773	2.887	2.746	−0.141	1199	3.989	4.024	+0.035	+0.001	0.84	12.548	12.355	−0.192
2	b0 f100 e10 s2	+0.0958	2.887	2.675	−0.212	1049	3.989	4.056	+0.067	+0.002	0.76	12.548	12.454	−0.094
3	b0 f100 e10 s3	+0.0715	2.887	2.827	−0.060	1094	3.989	4.066	+0.078	+0.002	0.83	12.548	12.360	−0.188
4	b0 f50 e10 s1	+0.0731	2.887	2.908	+0.021	1183	3.989	4.069	+0.081	+0.005	0.87	12.548	12.416	−0.131
5	b0 f50 e10 s2	+0.0765	2.887	2.987	+0.100	1318	3.989	4.052	+0.063	+0.004	0.85	12.548	12.454	−0.093
6	b0 f50 e10 s3	+0.0647	2.887	2.733	−0.154	1055	3.989	4.115	+0.127	+0.005	0.80	12.548	12.442	−0.105
7	b0 f25 e10 s1	+0.0643	2.887	2.829	−0.058	1230	3.989	4.009	+0.020	+0.003	0.86	12.548	12.331	−0.217
8	b0 f25 e10 s2	+0.0593	2.887	2.969	+0.082	1284	3.989	4.060	+0.071	+0.006	0.99	12.548	12.365	−0.182
9	b0 f25 e10 s3	+0.0672	2.887	2.746	−0.141	1237	3.989	4.023	+0.034	+0.005	0.91	12.548	12.434	−0.114
10	b0 f100 e5 s1	+0.0944	2.887	2.814	−0.073	1315	3.989	4.041	+0.052	+0.005	0.91	12.548	12.405	−0.143
11	b0 f100 e5 s2	+0.1008	2.887	2.888	+0.001	1279	3.989	4.028	+0.040	+0.003	0.87	12.548	12.409	−0.139
12	b0 f100 e5 s3	+0.1089	2.887	2.980	+0.093	1368	3.989	4.022	+0.033	+0.006	0.96	12.548	12.338	−0.210
13	b0 f100 e20 s1	+0.0693	2.887	2.808	−0.079	1024	3.989	4.039	+0.051	+0.007	0.82	12.548	12.395	−0.152
14	b0 f100 e20 s2	+0.0795	2.887	2.836	−0.051	989	3.989	4.048	+0.059	+0.003	0.77	12.548	12.359	−0.189
15	b0 f100 e20 s3	+0.1026	2.887	2.802	−0.085	1127	3.989	4.044	+0.056	+0.002	0.85	12.548	12.313	−0.235
16	b1 f100 e10 s1	+0.0677	2.946	2.775	−0.170	1132	4.007	4.028	+0.021	−0.001	0.79	12.203	12.380	+0.177
17	b1 f100 e10 s2	+0.0665	2.946	2.790	−0.156	1128	4.007	4.080	+0.073	+0.003	0.83	12.203	12.362	+0.159
18	b1 f100 e10 s3	+0.0648	2.946	2.666	−0.280	1108	4.007	4.033	+0.026	−0.001	0.80	12.203	12.419	+0.216
19	b1 f50 e10 s1	+0.0653	2.946	2.831	−0.115	1097	4.007	4.091	+0.084	+0.005	0.81	12.203	12.329	+0.126
20	b1 f50 e10 s2	+0.0667	2.946	2.793	−0.152	1061	4.007	4.095	+0.088	+0.002	0.84	12.203	12.352	+0.150
21	b1 f50 e10 s3	+0.0787	2.946	2.804	−0.141	1179	4.007	4.077	+0.070	+0.003	0.87	12.203	12.276	+0.073
22	b1 f25 e10 s1	+0.0483	2.946	2.887	−0.059	1127	4.007	4.036	+0.029	+0.001	0.81	12.203	12.227	+0.025
23	b1 f25 e10 s2	+0.0778	2.946	2.850	−0.095	1317	4.007	4.002	−0.004	+0.006	1.00	12.203	12.232	+0.029
24	b1 f25 e10 s3	+0.0631	2.946	2.835	−0.111	1292	4.007	4.028	+0.022	+0.004	0.84	12.203	12.417	+0.214
25	b1 f100 e5 s1	+0.0858	2.946	2.679	−0.267	1048	4.007	4.075	+0.068	+0.003	0.74	12.203	12.346	+0.143
26	b1 f100 e5 s2	+0.1180	2.946	2.851	−0.094	1204	4.007	4.071	+0.064	+0.004	0.85	12.203	12.371	+0.169
27	b1 f100 e5 s3	+0.1151	2.946	2.839	−0.107	1322	4.007	4.057	+0.050	+0.003	0.86	12.203	12.342	+0.140
28	b1 f100 e20 s1	+0.1023	2.946	2.851	−0.094	1102	4.007	4.054	+0.047	+0.000	0.77	12.203	12.412	+0.210
29	b1 f100 e20 s2	+0.0712	2.946	2.916	−0.029	1110	4.007	4.029	+0.022	+0.003	0.79	12.203	12.322	+0.120
30	b1 f100 e20 s3	+0.0708	2.946	2.920	−0.025	1101	4.007	4.056	+0.049	+0.002	0.83	12.203	12.323	+0.120
noise	old s0→old s1	–	2.887	2.946	+0.059	1118	3.989	4.007	+0.018	+0.001	0.78	12.548	12.203	−0.345
noise	old s1→old s0	–	2.946	2.887	−0.059	1119	4.007	3.989	−0.018	−0.001	0.73	12.203	12.548	+0.345

Old-pipeline Δ_2 is larger than isolated Δ_2 in 30 of 30 updates.

8.4 Entangled enhancement at three levels, per update

Y = harm at that level (shaded). L1 = any increase, L2 = above random variation ($\tau_2 = 0.112$, $\tau_3 = 0.018$ m), L3 = bootstrap 95 % interval above zero. EE = L1 for Δ_2 or Δ_3.

Table 13: Per-update classification (C_3 = ego)

		Old-pipeline training						Isolated training						
		Δ_2			Δ_3			Δ_2			Δ_3			
#	Update	L1	L2	L3	L1	L2	L3	L1	L2	L3	L1	L2	L3	EE (op)
1	b0 f100 e10 s1	Y									Y	Y		Y
2	b0 f100 e10 s2	Y			Y	Y					Y	Y	Y	Y
3	b0 f100 e10 s3	Y	Y		Y	Y					Y	Y	Y	Y
4	b0 f50 e10 s1	Y	Y					Y			Y	Y	Y	Y
5	b0 f50 e10 s2	Y	Y		Y	Y		Y			Y	Y	Y	Y
6	b0 f50 e10 s3	Y			Y	Y					Y	Y	Y	Y
7	b0 f25 e10 s1	Y	Y								Y	Y		Y
8	b0 f25 e10 s2	Y	Y	Y				Y			Y	Y		Y
9	b0 f25 e10 s3	Y	Y								Y	Y		Y
10	b0 f100 e5 s1	Y	Y								Y	Y		Y
11	b0 f100 e5 s2	Y	Y	Y				Y			Y	Y		Y
12	b0 f100 e5 s3	Y	Y	Y				Y			Y	Y		Y
13	b0 f100 e20 s1	Y									Y	Y		Y
14	b0 f100 e20 s2	Y	Y		Y	Y					Y	Y		Y
15	b0 f100 e20 s3	Y	Y		Y						Y	Y		Y
16	b1 f100 e10 s1										Y	Y		–
17	b1 f100 e10 s2				Y	Y					Y	Y	Y	Y
18	b1 f100 e10 s3				Y	Y					Y	Y		Y
19	b1 f50 e10 s1				Y	Y	Y				Y	Y	Y	Y
20	b1 f50 e10 s2				Y	Y					Y	Y	Y	Y
21	b1 f50 e10 s3				Y	Y					Y	Y		Y
22	b1 f25 e10 s1	Y									Y	Y		Y
23	b1 f25 e10 s2	Y												Y
24	b1 f25 e10 s3	Y			Y						Y	Y		Y
25	b1 f100 e5 s1				Y	Y					Y	Y	Y	Y
26	b1 f100 e5 s2	Y			Y						Y	Y	Y	Y
27	b1 f100 e5 s3	Y			Y						Y	Y		Y
28	b1 f100 e20 s1	Y			Y	Y					Y	Y		Y
29	b1 f100 e20 s2	Y			Y	Y					Y	Y		Y
30	b1 f100 e20 s3	Y			Y						Y	Y		Y
Total		23	11	3	18	13	1	5	0	0	29	29	10	29

9 Stage 3: bootstrap intervals for every update

Red = interval above zero (significant harm), green = interval below zero (significant improvement).

Table 14: Point value and 95 % bootstrap interval (2,000 resamples of 50 scenes), C_3 = ego, metres

#	Update	Δ_2 old-pipeline	Δ_3 old-pipeline	Δ_2 isolated	Δ_3 isolated
1	b0 f100 e10 s1	+0.049 [−0.198, +0.300]	−0.027 [−0.070, +0.018]	−0.141 [−0.370, +0.110]	+0.035 [−0.027, +0.111]
2	b0 f100 e10 s2	+0.062 [−0.166, +0.408]	+0.026 [−0.032, +0.094]	−0.212 [−0.396, +0.056]	+0.067 [+0.016, +0.124]
3	b0 f100 e10 s3	+0.139 [−0.096, +0.467]	+0.020 [−0.043, +0.080]	−0.060 [−0.244, +0.192]	+0.078 [+0.012, +0.158]
4	b0 f50 e10 s1	+0.184 [−0.020, +0.439]	−0.012 [−0.061, +0.035]	+0.021 [−0.140, +0.197]	+0.081 [+0.002, +0.200]
5	b0 f50 e10 s2	+0.192 [−0.010, +0.441]	+0.031 [−0.019, +0.086]	+0.100 [−0.094, +0.319]	+0.063 [+0.015, +0.119]
6	b0 f50 e10 s3	+0.098 [−0.133, +0.422]	+0.040 [−0.012, +0.095]	−0.154 [−0.362, +0.040]	+0.127 [+0.020, +0.297]
7	b0 f25 e10 s1	+0.233 [−0.032, +0.516]	−0.030 [−0.076, +0.011]	−0.058 [−0.314, +0.180]	+0.020 [−0.028, +0.073]
8	b0 f25 e10 s2	+0.284 [+0.079, +0.544]	−0.042 [−0.089, +0.001]	+0.082 [−0.108, +0.321]	+0.071 [−0.009, +0.172]
9	b0 f25 e10 s3	+0.157 [−0.026, +0.367]	−0.046 [−0.102, +0.003]	−0.141 [−0.327, +0.020]	+0.034 [−0.038, +0.126]
10	b0 f100 e5 s1	+0.182 [−0.033, +0.397]	−0.024 [−0.067, +0.023]	−0.073 [−0.290, +0.143]	+0.052 [−0.027, +0.150]
11	b0 f100 e5 s2	+0.284 [+0.047, +0.549]	−0.006 [−0.049, +0.035]	+0.001 [−0.227, +0.249]	+0.040 [−0.015, +0.105]
12	b0 f100 e5 s3	+0.266 [+0.033, +0.545]	−0.042 [−0.092, +0.006]	+0.093 [−0.134, +0.347]	+0.033 [−0.041, +0.127]
13	b0 f100 e20 s1	+0.107 [−0.150, +0.419]	−0.000 [−0.057, +0.052]	−0.079 [−0.285, +0.136]	+0.051 [−0.018, +0.128]
14	b0 f100 e20 s2	+0.127 [−0.113, +0.482]	+0.025 [−0.053, +0.104]	−0.051 [−0.316, +0.290]	+0.059 [−0.001, +0.121]
15	b0 f100 e20 s3	+0.115 [−0.111, +0.384]	+0.001 [−0.065, +0.065]	−0.085 [−0.289, +0.142]	+0.056 [−0.015, +0.139]
16	b1 f100 e10 s1	−0.054 [−0.238, +0.194]	−0.001 [−0.064, +0.069]	−0.170 [−0.355, +0.028]	+0.021 [−0.023, +0.071]
17	b1 f100 e10 s2	−0.081 [−0.352, +0.232]	+0.026 [−0.036, +0.089]	−0.156 [−0.445, +0.119]	+0.073 [+0.014, +0.147]
18	b1 f100 e10 s3	−0.187 [−0.376, +0.044]	+0.023 [−0.036, +0.084]	−0.280 [−0.484, −0.082]	+0.026 [−0.029, +0.085]
19	b1 f50 e10 s1	−0.030 [−0.190, +0.204]	+0.072 [+0.009, +0.139]	−0.115 [−0.288, +0.074]	+0.084 [+0.001, +0.186]
20	b1 f50 e10 s2	−0.135 [−0.329, +0.081]	+0.054 [−0.020, +0.131]	−0.152 [−0.363, +0.038]	+0.088 [+0.011, +0.189]
21	b1 f50 e10 s3	−0.089 [−0.266, +0.087]	+0.037 [−0.015, +0.089]	−0.141 [−0.341, +0.045]	+0.070 [−0.015, +0.181]
22	b1 f25 e10 s1	+0.090 [−0.062, +0.302]	−0.001 [−0.050, +0.046]	−0.059 [−0.234, +0.109]	+0.029 [−0.024, +0.088]
23	b1 f25 e10 s2	+0.081 [−0.082, +0.250]	−0.042 [−0.101, +0.019]	−0.095 [−0.303, +0.089]	−0.004 [−0.058, +0.056]
24	b1 f25 e10 s3	+0.082 [−0.102, +0.278]	+0.010 [−0.050, +0.069]	−0.111 [−0.323, +0.072]	+0.022 [−0.029, +0.079]
25	b1 f100 e5 s1	−0.025 [−0.203, +0.256]	+0.033 [−0.016, +0.087]	−0.267 [−0.450, −0.075]	+0.068 [+0.011, +0.138]
26	b1 f100 e5 s2	+0.040 [−0.116, +0.262]	+0.003 [−0.038, +0.047]	−0.094 [−0.239, +0.054]	+0.064 [+0.001, +0.137]
27	b1 f100 e5 s3	+0.052 [−0.113, +0.266]	+0.015 [−0.023, +0.055]	−0.107 [−0.283, +0.065]	+0.050 [−0.001, +0.109]
28	b1 f100 e20 s1	+0.003 [−0.180, +0.278]	+0.049 [−0.012, +0.122]	−0.094 [−0.296, +0.105]	+0.047 [−0.017, +0.123]
29	b1 f100 e20 s2	+0.065 [−0.134, +0.389]	+0.021 [−0.062, +0.100]	−0.029 [−0.209, +0.185]	+0.022 [−0.029, +0.078]
30	b1 f100 e20 s3	+0.073 [−0.129, +0.374]	+0.018 [−0.062, +0.101]	−0.025 [−0.208, +0.155]	+0.049 [−0.014, +0.119]
noise	old s0→old s1	+0.112 [−0.099, +0.314]	−0.013 [−0.049, +0.024]	+0.059 [−0.174, +0.318]	+0.018 [−0.013, +0.049]
noise	old s1→old s0	−0.098 [−0.347, +0.153]	+0.001 [−0.034, +0.039]	−0.059 [−0.318, +0.174]	−0.018 [−0.049, +0.013]

10 H2: output change without ground truth, every update

Table 15: Nine label-free measures per update (old C_1 vs updated C_1 on Singapore 2), with δ_1 and the old-pipeline Δ_2

#	Update	δ_1	Δ_2 op	obj/img JS	mean obj/img	type JS	conf W	conf KS	fwd W (m)	lat W (m)	only-one	C2 move W	C2 shift (m)
1	b0 f100 e10 s1	+0.0773	+0.049	0.0097	2.84/3.13	0.0335	0.0211	0.062	11.43	2.50	0.516	0.291	4.49
2	b0 f100 e10 s2	+0.0958	+0.062	0.0132	2.96/2.83	0.0340	0.0127	0.052	87.28	9.90	0.494	0.317	4.54
3	b0 f100 e10 s3	+0.0715	+0.139	0.0068	2.89/3.01	0.0370	0.0153	0.068	10.21	3.62	0.530	0.416	4.51
4	b0 f50 e10 s1	+0.0731	+0.184	0.0200	2.79/2.82	0.0387	0.0201	0.062	36.48	6.44	0.498	0.292	4.41
5	b0 f50 e10 s2	+0.0765	+0.192	0.0186	2.81/3.19	0.0306	0.0114	0.046	41.74	5.22	0.506	0.522	4.84
6	b0 f50 e10 s3	+0.0647	+0.098	0.0152	2.97/2.69	0.0310	0.0290	0.075	13.16	5.25	0.528	0.379	4.36
7	b0 f25 e10 s1	+0.0643	+0.233	0.0174	2.81/3.05	0.0307	0.0155	0.058	9.07	3.03	0.511	0.564	4.59
8	b0 f25 e10 s2	+0.0593	+0.284	0.0292	2.69/3.06	0.0209	0.0113	0.043	36.90	7.02	0.535	0.383	4.88
9	b0 f25 e10 s3	+0.0672	+0.157	0.0321	2.69/2.97	0.0208	0.0087	0.040	161.94	37.70	0.474	0.197	4.32
10	b0 f100 e5 s1	+0.0944	+0.182	0.0184	2.84/3.26	0.0312	0.0269	0.066	61.07	10.24	0.492	0.214	4.59
11	b0 f100 e5 s2	+0.1008	+0.284	0.0110	2.84/3.15	0.0334	0.0163	0.046	7.55	3.03	0.510	0.338	4.42
12	b0 f100 e5 s3	+0.1089	+0.266	0.0332	2.70/3.27	0.0350	0.0135	0.050	20.39	6.80	0.529	0.481	5.24
13	b0 f100 e20 s1	+0.0693	+0.107	0.0108	2.91/2.69	0.0308	0.0483	0.118	87.45	8.83	0.535	0.447	4.87
14	b0 f100 e20 s2	+0.0795	+0.127	0.0101	2.86/2.66	0.0384	0.0492	0.126	18.85	6.82	0.541	0.790	4.66
15	b0 f100 e20 s3	+0.1026	+0.115	0.0103	2.83/2.90	0.0413	0.0368	0.103	27.71	8.03	0.545	0.612	4.48
16	b1 f100 e10 s1	+0.0677	−0.054	0.0182	2.98/2.74	0.0367	0.0350	0.075	11.02	5.26	0.511	0.358	1.62
17	b1 f100 e10 s2	+0.0665	−0.081	0.0173	3.00/2.86	0.0359	0.0305	0.075	25.42	4.60	0.514	0.307	1.72
18	b1 f100 e10 s3	+0.0648	−0.187	0.0132	2.97/2.84	0.0378	0.0286	0.064	16.05	2.38	0.533	0.421	1.70
19	b1 f50 e10 s1	+0.0653	−0.030	0.0198	2.98/2.76	0.0398	0.0182	0.051	17.96	6.19	0.510	0.421	1.74
20	b1 f50 e10 s2	+0.0667	−0.135	0.0343	3.04/2.73	0.0361	0.0034	0.017	28.09	5.53	0.533	0.442	1.82
21	b1 f50 e10 s3	+0.0787	−0.089	0.0178	3.02/2.93	0.0427	0.0238	0.061	22.27	4.54	0.508	0.452	1.70
22	b1 f25 e10 s1	+0.0483	+0.090	0.0133	2.97/2.75	0.0314	0.0171	0.051	21.87	3.88	0.503	0.348	1.61
23	b1 f25 e10 s2	+0.0778	+0.081	0.0214	2.74/3.10	0.0392	0.0125	0.046	57.17	7.48	0.541	0.562	1.82
24	b1 f25 e10 s3	+0.0631	+0.082	0.0090	2.93/3.13	0.0301	0.0114	0.034	48.05	7.35	0.473	0.386	1.59
25	b1 f100 e5 s1	+0.0858	−0.025	0.0409	3.07/2.62	0.0387	0.0265	0.063	16.57	3.57	0.490	0.166	1.62
26	b1 f100 e5 s2	+0.1180	+0.040	0.0079	2.92/2.94	0.0359	0.0281	0.068	18.82	6.79	0.507	0.390	1.82
27	b1 f100 e5 s3	+0.1151	+0.052	0.0084	2.94/3.18	0.0342	0.0273	0.062	14.49	3.58	0.514	0.367	1.60
28	b1 f100 e20 s1	+0.1023	+0.003	0.0315	3.01/2.71	0.0428	0.0427	0.097	23.79	5.00	0.534	0.614	1.72
29	b1 f100 e20 s2	+0.0712	+0.065	0.0226	2.95/2.72	0.0434	0.0355	0.084	17.41	4.22	0.542	0.591	1.83
30	b1 f100 e20 s3	+0.0708	+0.073	0.0208	2.97/2.89	0.0395	0.0384	0.088	22.43	4.44	0.537	0.672	1.65

Table 16: How well each measure separates harmful updates ($\Delta_2 > 0$, old-pipeline)

	All 30		Base s0 (15)		Base s1 (15)	
Measure	ρ	AUROC	ρ	AUROC	ρ	AUROC
δ_1 (needs ground truth)	+0.04	0.66	−0.03	–	−0.01	0.64
objects per image JS	−0.01	0.32	+0.58	–	−0.14	0.38
type mix JS	−0.51	0.26	−0.26	–	−0.20	0.45
confidence Wasserstein	−0.29	0.46	−0.45	–	−0.10	0.57
confidence KS	−0.21	0.52	−0.56	–	−0.09	0.57
forward position Wasserstein	+0.13	0.63	−0.19	–	+0.25	0.64
lateral position Wasserstein	+0.26	0.68	−0.16	–	+0.17	0.59
only-one-version fraction	−0.02	0.58	−0.10	–	−0.04	0.62
C2 predicted movement Wasserstein	+0.02	0.60	+0.13	–	+0.08	0.70
C2 shift of the same object	+0.68	0.81	+0.29	–	−0.28	0.45

In base s0 all 15 updates have $\Delta_2 > 0$, so the AUROC within base s0 is undefined (–).

11 Summary of the counts

Table 17: Updates (out of 30) with an increase of the error

	Prediction Δ_2		Planning Δ_3 (ego)	
Level	op	iso	op	iso
L1 any increase	23	5	18	29
L2 above random variation	11	0	13	29
L3 bootstrap interval above zero	3	0	1	10

- Every one of the 30 updates improved C_1 (δ_1 from +0.0483 to +0.1180).
- Entangled enhancement by the simple rule (L1, Δ_2 or Δ_3 after old-pipeline training): 29 of 30.
- New-pipeline training (Stage 1b, 2 updates): C_2 2.748/2.606 m → 2.389/2.446 m; C_3 ego 4.143/4.000 m → 3.705/3.697 m.